Partho Ghosh

Organized — Passionate — Detail-Oriented — Enthusiastic About AI & Technologies

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Career Objective

A third-year PhD researcher in Biomedical Engineering specializing in AI-driven medical diagnostics and intelligent systems, passionate about pioneering next-generation artificial intelligence solutions. Dedicated to transforming academic breakthroughs into real-world impact through innovations in healthcare imaging, adaptive AI methodologies, and scalable intelligent frameworks.

Education

2024 - Present ◇ **Ph.D. in Biomedical Engineering**, The University of Iowa

- Research Focus: Pulmonary image analysis, medical computer vision
- Academic Advisor: Dr. Sarah E. Gerard
- GPA: 4.00/4.00

2017 - 2022 ◇ **B.S. in Electrical & Electronic Engineering**, BUET, Bangladesh

- Major: Communication and Signal Processing
- CGPA: 3.38/4.00

Positions and Employment

Jan 2024 – Present ◇ **Graduate Research Assistant**, PICV Lab, University of Iowa

- Conduct medical image analysis using deep learning (CT, MRI)
- Supervisor: Dr. Sarah E. Gerard

Jan 2023 – Nov 2023 ◇ **Machine Learning Engineer**, adorsho praniSheba Ltd., Bangladesh

- Improved cattle identification algorithms and heat detection systems
- Developed weight measurement systems using mobile camera images

July 2019 – Dec 2023 ◇ **Research Assistant**, mHealth Lab, BUET, Bangladesh

- Led BUET-Synapticans in international competitions
- Focus on domain adaptation in video activity recognition

Jan 2019 – July 2019 ◇ **Web Developer**, Suffix IT Ltd., Bangladesh

- Full-stack developer for the project of Institute of Engineers' Bangladesh (IEB) website

Awards and Honors

2025 ◇ **Graduate Student Senate (GSS) Travel Funds Award**, University of Iowa

2025 ◇ **Krishnan B. Chandran Graduate Student Travel Awards**, University of Iowa

2022 ◇ **Top Data Scientist**, Robi Datathon 2.0

2021 ◇ **2nd Runner-up**, INNOVETIA-2021

2021 ◇ **5th Position**, IEEE Signal Processing Cup-2020

2020 ◇ **Champion**, IEEE Video and Image Processing Cup-2020

2020 ◇ **7th Position**, Stall Catchers Alzheimer's research Competition

2019 ◇ **2nd Runner-up**, IEEE Video and Image Processing Cup-2019

Research Experience

Nov 2024 – Present ◇ **Lung Foundation Model Development**

- Investigating Semi-supervised Training frameworks to build models that have understanding of the lungs

Aug 2024 – Dec 2024 ◇ **Lung Segmentation in UTE MRI**

- Leveraging CT data for MRI segmentation; accepted at ISBI 2025

Feb 2024 – Present ◇ **Airway Segmentation in Fibrotic Diseased Patient**

- Tackling honeycomb problem in airway segmentation

Mar 2023 – Feb 2024 ◇ **Domain Generalization in Office Activity Recognition**

- Proposed adapting layers to address environmental changes in activity recognition

Dec 2021 – Oct 2022 ◇ **Clogged Blood Vessel Detection**

- Sequence-agnostic multimodal approach; published in ICASSP 2023 Workshops

Jun 2018 – Dec 2018 ◇ **Food Recognition & Nutritional Value Estimation**

- Developed a CNN-based system for food recognition and calorie calculation

Publications

1. **Partho Ghosh**, Keegan R. Staab, Jonathan L. Percy, Abhilash Kizhakke Puliyakote, Marrissa McIntosh, Andrew Hahn, Sean B. Fain and Sarah E. Gerard, "Multimodal Learning for Lung Segmentation: Enhancing UTE MRI Segmentation with CT datasets," 2025 IEEE International Symposium on Biomedical Imaging (ISBI), Houston, TX, USA 2025, **DOI:** 10.1109/ISBI60581.2025.10980692

2. **Partho Ghosh**, Md. Abrar Istiak, Mir Sayeed Mohammad, Swapnil Saha and Uday Kamal, "A Sequence Agnostic Multimodal Pre-Processing for Clogged Blood Vessel Detection in Alzheimer's Diagnosis," 2023 IEEE International Conference on Acoustics, Speech, and Signal Processing Workshops (ICASSPW), Rhodes Island, Greece, 2023, pp. 1-5, **DOI:** 10.1109/ICASSPW59220.2023.10193405.

3. **Partho Ghosh**, Md Abrar Istiak, Nayeeb Rashid, Ahsan Habib Akash, Ridwan Abrar, Ankan Ghosh Dastider, Asif Shahriyar Sushmit, and Taufiq Hasan. "Activity Classification from First-Person Office Videos with Visual Privacy Protection." In The Fourth Industrial Revolution and Beyond: Select Proceedings of IC4IR+, pp. 157-169. Singapore: Springer Nature Singapore, 2023. **DOI:** 10.1007/978-981-19-8032-9_12

arxiv

1. **Partho Ghosh**, Raisa Bentay Hossain, Mohammad Zunaed and Taufiq Hasan, "Domain Generalization for Improved Human Activity Recognition in Office Space Videos Using Adaptive Pre-processing." arXiv preprint arXiv:2503.12678 (2025). **DOI:** 10.48550/arXiv.2503.12678

2. **Partho Ghosh**, and Md Aynal Haque. "ResDTA: Predicting Drug-Target Binding Affinity Using Residual Skip Connections." 2023. arXiv preprint **DOI:** 10.48550/arXiv.2303.11434

3. Asif Shahriyar Sushmit, **Partho Ghosh**, Md Abrar Istiak, Nayeeb Rashid, Ahsan Habib Akash, and Taufiq Hasan. "Segcodenet: Color-coded segmentation masks for activity detection from wearable cameras." 2020. arXiv preprint **DOI:** 10.48550/arXiv.2008.08452

Presentation

1. **Partho Ghosh**, Keegan R. Staab, Jonathan L. Percy, Abhilash Kizhakke Puliyakote, Marrissa McIntosh, Andrew Hahn, Sean B. Fain and Sarah E. Gerard, "Roughness-Enforced Causal Augmentation for Generalizable Lung Segmentation" 2026 Research Open House(ROH), University of Iowa, Iowa City, IA, USA.

2. **Partho Ghosh**, Keegan R. Staab, Jonathan L. Percy, Abhilash Kizhakke Puliyakote, Marrissa McIntosh, Andrew Hahn, Sean B. Fain and Sarah E. Gerard, "Multimodal Learning for Lung Segmentation: Enhancing UTE MRI Segmentation with CT datasets," 2025 IEEE International Symposium on Biomedical Imaging (ISBI), Houston, TX, USA 2025, **DOI:** 10.1109/ISBI60581.2025.10980692

3. **Partho Ghosh**, Keegan R. Staab, Jonathan L. Percy, Abhilash Kizhakke Puliyakote, Marrissa McIntosh, Andrew Hahn, Sean B. Fain and Sarah E. Gerard, "Multimodal Learning for Lung Segmentation: Enhancing UTE MRI Segmentation with CT datasets," 2025 Research Open House(ROH), University of Iowa, Iowa City, IA, USA.

4. **Partho Ghosh** and Sarah E. Gerard, "Deep Learning for Automated Airway Segmentation in Chest CT Scans," 2024 Research Open House(ROH), University of Iowa, Iowa City, IA, USA.

Relevant Courses

- Linear Algebra, Probability & Statistics
- Digital Image Processing, Signal Processing, Medical Imaging Physics, Applied Machine Learning
- Deep Learning in Medical Imaging, Large Language Models

References

Dr. Sarah E. Gerard
Assistant Professor, Biomedical Engineering, University of Iowa
Email: sarah-gerard@uiowa.edu

Dr. Taufiq Hasan
Associate Professor, Biomedical Engineering, BUET
Email: taufiq@bme.buet.ac.bd